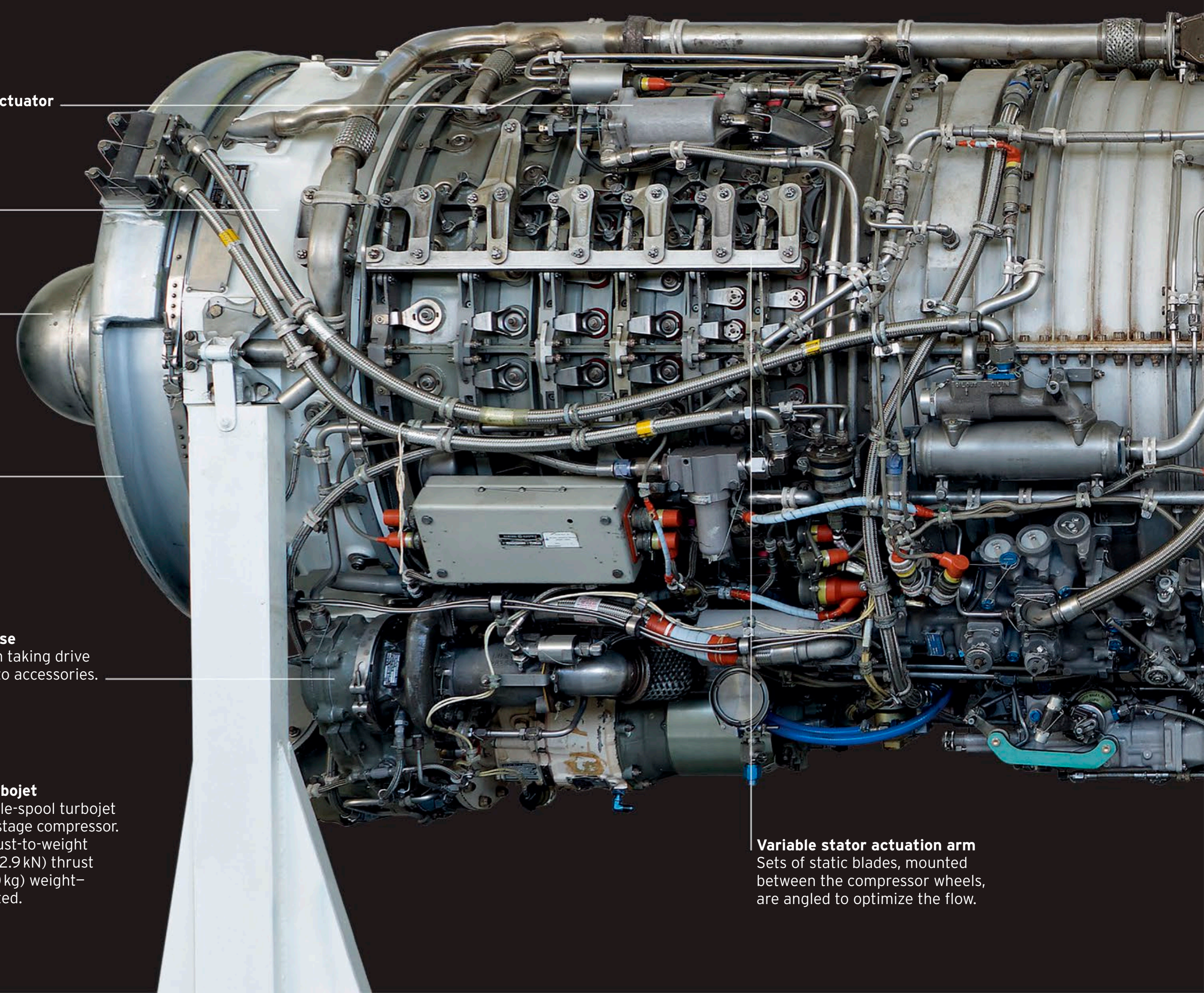
General Electric J79

At the time of its introduction, the Mach 2-capable General Electric J79 was the most advanced turbojet ever designed. It was used to power iconic jet fighters including the F-104 Starfighter, the F-4 Phantom, and some F-16 Fighting Falcons. The adaptable powerplant found use in the B-58 Hustler bomber and, in a civilian version, the Convair 880/990 airliner family.

Variable blades
This engine's novel arrangement of variable stator blades enable it to generate twin-spool-like power at a much lower weight. Variable blades were an important development in engine design.

RECORD BREAKER

General Electric's J79 program began in 1954 as a development of the earlier J73. The new engine was designed to be capable of Mach 2 and made its test flight (in the bomb bay of a B-45 Tornado) on May 20, 1955. The powerplant would go on to set the world altitude record—91,249 ft (27,812 m)—and speed record—more than 1,400 mph (2,253 km/h)—in an F-104 Starfighter. More than 17,000 J79s were built and around 1,300 are still in service, with many expected to remain flying beyond 2020.



Transfer gear case
Houses gear train taking drive from main shaft to accessories.

Single-spool turbojet
The J79 is a single-spool turbojet engine with a 17-stage compressor. The engine's thrust-to-weight ratio—11,906 lb (52.9 kN) thrust to 3,850 lb (1,750 kg) weight—was unprecedented.

Variable stator actuation arm
Sets of static blades, mounted between the compressor wheels, are angled to optimize the flow.